Hao (Mark) Chen

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RESEARCH INTERESTS

My research focuses on democratizing AI through efficient ML algorithms (e.g., LLM quantization, speculative decoding, model merging, expert skipping) and systems (e.g., memory management, request scheduling, heterogeneous computing).

EDUCATION

Imperial College London
PhD in Computing — *Sep 2024 – June 2028 (expected)*
Advisor: Prof. Wayne Luk, Dr. Hongxiang Fan

Imperial College London
MEng in Computing — *Oct 2020 – June 2024*
Advisor: Prof. Wayne Luk, Dr. Hongxiang Fan

PUBLICATIONS & MANUSCRIPTS

Conference Papers

- **Hao Mark Chen**, Zhiwen Mo, Guanxi Lu, Shuang Liang, Lingxiao Ma, Wayne Luk, Hongxiang Fan. Democratizing Agentic AI with Fast Test-Time Scaling on the Edge. Accepted at *2026 International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. [Paper]
- **Hao Mark Chen**, Guanxi Lu, Yasuyuki Okoshi, Zhiwen Mo, Masato Motomura, Hongxiang Fan. Rethinking Optimal Verification Granularity for Compute-Efficient Test-Time Scaling. Accepted at *2025 Conference on Neural Information Processing Systems (NeurIPS)*. [Paper] [Code]
- **Hao (Mark) Chen**, Wayne Luk, Ka Fai Cedric Yiu, Rui Li, Konstantin Mishchenko, Stylianos I Venieris, Hongxiang Fan. Hardware-Aware Parallel Prompt Decoding for Memory-Efficient Acceleration of LLM Inference. Accepted at *2025 Empirical Methods in Natural Language Processing (EMNLP)*. [Paper] [Code]
- **Hao Mark Chen**, Shell Xu Hu, Wayne Luk, Timothy Hospedales, Hongxiang Fan. FW-Merging: Scaling model merging with frank-wolfe optimization. Accepted at *2025 International Conference on Computer Vision (ICCV)*. [Paper] [Code]
- **Hao Mark Chen**, Fuwen Tan, Alexandros Kouris, Royson Lee, Hongxiang Fan, Stylianos I Venieris. Progressive Mixed-Precision Decoding for Efficient LLM Inference. Accepted at *2025 International Conference on Learning Representations (ICLR)*. [Paper] [Code]
- **Hao Mark Chen**, Zehuan Zhang, Wanru Zhao, Nicholas Lane, Hongxiang Fan. Advancing AI-assisted Hardware Design with Hierarchical Decentralized Training and Personalized Inference-Time Optimization. [Paper]
- Guanxi Lu, **Hao Mark Chen**, Yuto Karashima, Zhican Wang, Daichi Fujiki, Hongxiang Fan. AdaBlock-dLLM: Semantic-Aware Diffusion LLM Inference via Adaptive Block Size. [Paper]
- Charlie Campbell, **Hao Mark Chen**, Wayne Luk, Hongxiang Fan. Enhancing LLM-based Quantum Code Generation with Multi-Agent Optimization and Quantum Error Correction. Accepted at *2025 Design Automation Conference (DAC)*. [Paper]
- Jiahao Gai, **Hao (Mark) Chen**, Zhican Wang, Hongyu Zhou, Wanru Zhao, Nicholas Lane, Hongxiang Fan. Exploring Code Language Models for Automated HLS-based Hardware Generation: Benchmark, Infrastructure and Analysis. Accepted at *2025 Asia and South Pacific Design Automation Conference (ASP-DAC)*. [Paper]
- Zehuan Zhang, Hongxiang Fan, **Hao Chen**, Lukasz Dudziak, Wayne Luk. Hardware-Aware Neural Dropout Search for Reliable Uncertainty Prediction on FPGA. Accepted at *2024 Design Automation Conference (DAC)*. [Paper]

- **Hao (Mark) Chen**, Taowen Liu, Songyun Hu, Leyang Yu, Yiqi Li, Sihan Tao, Jacqueline Lee, Ahmed E. Fetit. Web-Based AI System for Medical Image Segmentation. Accepted at *2023 Conference on Medical Image Understanding and Analysis (MIUA)*. [Paper]
- Hongxiang Fan, **Hao (Mark) Chen**, Liam Castelli, Zhiqiang Que, He Li, Kenneth Long, Wayne Luk. When Monte-Carlo Dropout Meets Multi-Exit: Optimizing Bayesian Neural Networks on FPGA. Accepted at *2023 Design Automation Conference (DAC)*. [Paper]

Journal Papers

- **Hao (Mark) Chen**, Liam Castelli, Martin Ferianc, Shuanglong Liu, Wayne Luk, Hongxiang Fan. Algorithm and Hardware Co-Design for Multi-Exit Dropout-based Bayesian Neural Networks. [Paper]

INDUSTRIAL EXPERIENCE

Samsung AI Center
Research Intern, Cambridge, UK *July 2025 – December 2025*
Worked on diffusion language model.

Samsung AI Center
Research Intern, Cambridge, UK *July 2024 – December 2024*
Worked on efficient LLM inference involving quantization, leading to a publication at *ICLR 2025*.

Qube RT
Quantative Technologist Intern, UK *April 2023 – Sep 2023*
Developed C++ performance monitoring system for low-latency trading platform.

Huawei Technologies Research & Development
Graphics Modelling Intern, UK *March 2022 – Sep 2022*
Worked on automatic GPU API generation.

Ampere Computing Shanghai, China
Java Software Developer *June 2021 – Sep 2021*
Developed a log search engine for Jenkins based on Lucene Search.

AWARDS

- **President's PhD scholarships** 2024 – 2028
- **Winton Capital Applied Undergraduate Project Computing Prize** 2024
- **Governors' MEng Prize in Computing** (the best overall performance in MEng programme) 2024
- **Dean's List (Year 4)**, Imperial College London 2024
- **Dean's List (Year 3)**, Imperial College London 2023
- **Dean's List (Year 2)**, Imperial College London 2022
- **Dean's List (Year 1)**, Imperial College London 2021
- **G-Research Ltd Prize**, G-Research Ltd, 2021
- **Singapore SM1 Scholarship**, Ministry of Education Singapore 2015 – 2019

SKILLS

- **Programming:** C++, C, Python, Scala, Java, Swift, Haskell, Bash
- **Tools:** GCC, Jenkins, Github, Docker, Heroku, AWS
- **Framework:** PyTorch, Keras, ODB, Kitura, Lucene, Jinja2
- **Language:** GRE - 333/340 + 5/6 (Verbal 163, Quantitative 170, Analytical Writing 5); TOEFL iBT 113/120 (Reading 30, Listening 29, Speaking 26, Writing 28)